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Box erecting apparatus and method

Abstract

An apparatus and method for expanding a box blank into a box, the apparatus comprising an arm assembly, a controller, a camera and a box blank conveyor. The arm assembly includes a folding arm having a position in a first direction and a rotational angle controlled by the controller based on a position of a feature of the box blank in the field of view of the camera. The camera captures images of the box blank which are used to position the arm assembly and to evaluate the need to reject a box blank.

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Background/Summary

BACKGROUND

(1) Shipping boxes keep customer order items together as well as protect the contents of the box. Corrugated cardboard boxes are especially useful because flat box blanks are convenient to ship to a manufacturing or distribution business. At the business site the flat box blank may be expanded, or erected, by folding the box blank at score lines, or creases, to form a box. Panels of the box blank are separated by score lines. Once expanded into a box, product may be inserted into the box for storage and/or shipping. Box erecting, or expanding, machines may automate the folding of a flat box blank for use in high volume production or distribution situations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a perspective view of a box erecting machine, in accordance with some embodiments.
- (3) FIG. 2 is a top view of the box erecting machine of FIG. 1, in accordance with some embodiments.
- (4) FIG. 3 is a top view of an arm assembly, in accordance with some embodiments.
- (5) FIG. 4 is a side view of the box erecting machine of FIG. 1, in accordance with some embodiments.
- (6) FIG. 5A is a perspective view of an expanding apparatus of a box erecting machine of FIG. 1, showing a field of view of a CCD camera, in accordance with some embodiments.
- (7) FIG. 5B is a view of a flat box blank seen by the CCD camera prior to box erection, in accordance with some embodiments.
- (8) FIG. 5C is a view of a folding arm of a box erecting machine of FIG. 1, in accordance with some embodiments.
- (9) FIG. 6A is a top view of an expanding apparatus of the box erecting machine of FIG. 1, prior to erecting a box, in accordance with some embodiments.

- (10) FIG. 6B is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with a pick apparatus advanced to pick the flat box blank, in accordance with some embodiments.
- (11) FIG. 6C is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with a folding arm contacting the box blank, in accordance with some embodiments.
- (12) FIG. 6D is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with a box partially erected from the box blank, in accordance with some embodiments.
- (13) FIG. 6E is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with the box erected, in accordance with some embodiments.
- (14) FIG. 6F is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with the box ready to be pushed onto an exit conveyor, in accordance with some embodiments.
- (15) FIG. 6G is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with the box being pushed onto the exit conveyor, in accordance with some embodiments.
- (16) FIG. 6H is a top view of the expanding apparatus of the box erecting machine of FIG. 1 with the arm assembly in position to expand the next box blank.
- (17) FIG. 7A is a perspective view of the expanding apparatus of FIGS. 6A-6F, in accordance with some embodiments.
- (18) FIG. 7B illustrates comparison of score line offset to score line offset thresholds in accordance with some embodiments.
- (19) FIG. 7C is a side view of vacuum suction cups of the expanding apparatus of FIGS. 6A-6F, in accordance with some embodiments.
- (20) FIG. 7D is a side view of the expanding apparatus of FIGS. 6A-6F, showing rejection of a flat box blank, in accordance with some embodiments.
- (21) FIG. 8 is a flow chart of a setup procedure for the box erecting machine of FIG. 1 for operation, in accordance with some embodiments.
- (22) FIG. 9 is a flow chart of operation of the box erecting machine of FIG. 1, in accordance with some embodiments.
- (23) FIG. 10 is a flow chart of operation of a box ejection procedure for the box erecting machine of FIG. 1, in accordance with some embodiments.

DETAILED DESCRIPTION

- (24) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (25) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (26) Box erecting, or expanding, machines automatically fold box blanks at score lines between panels of a cardboard box. These machines are suitable for manufacturing or distribution centers that ship large volumes of product in cardboard boxes. Automatic functions such as adjusting to a

variety of box sizes can reduce injury to machine operators and increase productivity.

(27) FIG. 1 shows a box erecting machine **100** in perspective view, in accordance with an embodiment of the present disclosure. A set of three directional axes X, Y, and Z are shown to facilitate description of FIGS. 1-7C. The box erecting machine **100** includes a box feed hopper, or box blank cartridge **102** which holds multiple flat boxes, or box blanks **104**, and feeds the box blanks **104** to an expanding apparatus **106** one at a time. A first box blank **121** is the one of the box blanks **104** staged in the box blank cartridge **102** and is closest to the expanding apparatus **106**. The first box blank **121** is positioned to be pulled into the expanding apparatus **106** where it will be expanded or opened up to form a usable box. In accordance with the illustrated embodiment, first box blank **121** is pulled into expanding apparatus **106** using vacuum suction cups **108**. As a box blank **104** is individually pulled into the expanding apparatus **106**, forces are applied to panels and flaps of the box blanks **104** such that the box blanks **104** bend at scores lines and the box blanks **104** expand to form a container, or box **110**. The box **110** exits the box erecting machine **100** on an exit conveyor **112**. Operation of the box erecting machine **100** is controlled by a machine controller **113** having an operator interface **115**.

(28) The box blank cartridge **102** includes a cartridge follower **114**, a right box blank guide **116** and a left box blank guide **118**. The right and left box blank guides **116** and **118** keep the box blanks **104** aligned with the expander apparatus in the box blank cartridge **102**. In operation, an operator places box blanks **104** into the box blank cartridge **102** with the box blanks **104** upright between the right box blank guide **116** and the left box blank guide **118**. Panels of the box blanks **104** face either the expanding apparatus **106** or the cartridge follower **114**. The cartridge follower **114** moves to contact panels of the last box blank **119** in the box blank cartridge **102**. The box blank cartridge **102** is sloped downward, toward the expanding apparatus **106**, allowing gravity to assist movement of the box blanks **104** toward the expanding apparatus **106**. The cartridge follower **114** maintains contact with the last box blank **119** in the box blank cartridge **102**, keeping the box blanks **104** upright. Various sizes of box blanks **104** may be accommodated by the box blank cartridge **102** by moving the right box blank guide **116** toward, or away from, the left box blank guide **118** to accommodate a width of the box blanks **104**.

(29) FIG. 2 is a top view of the box erecting machine **100**. The box blanks **104** and the box **110** (see FIG. 1) are not shown. The box erecting machine **100** includes a box blank conveyor **127** for moving the first box blank **121** into the expanding apparatus **106**. The box blank conveyor **127** includes vacuum suction cups **108** coupled to a gantry **120** which moves along gantry rails **122** and **124**. Movement of gantry **120** along gantry rails **122** and **124** causes the vacuum suction cups **108** to move parallel to the X axis. A position of the vacuum suction cups **108** is determined by a gantry motor **126** coupled between the gantry **120** and the gantry rail **124**. The gantry motor **126** is controlled by the machine controller **113**. The expanding apparatus **106** also includes an arm assembly **128** and an imaging device, such as a camera **130**. The camera **130** is pointed toward the box blank cartridge **102**, so that the imaging portion of the camera can capture an image of a box blank about to enter the expanding apparatus **106**, also known as the next box blank. The expanding apparatus **106** may also have a gate latch **132** located at an edge of the expanding apparatus **106** adjoining the box blank cartridge **102**. The gate latch **132**, which may have a solenoid actuator, may be positioned to prevent the box blanks **104** from moving from the box blank cartridge **102** to the expanding apparatus **106** and is controlled by the machine controller **113**.

(30) FIG. 3 is a top view of an arm assembly **128** in accordance with an embodiment of the present disclosure. Arm assembly **128** includes a push arm **134** and a folding arm **136** coupled to a base **138**. The base **138** has a surface **139** to which components of the arm assembly **128** are coupled. In the illustrated embodiment, the push arm **134** is rigidly coupled to the surface **139** of the base **138**. A side or face **140** of the push arm **134** faces the exit conveyor **112**, along the Y axis. The folding arm **136** is coupled to the base **138** by a folding arm pivot joint or a folding arm rotation joint **142** allowing rotation of the folding arm **136** parallel to an X-Y plane (horizontal plane) and around the

Z-axis (vertical axis) in FIG. 3. A side or face **144** of the folding arm **136** faces the box blank cartridge **102** at a folding arm angle **146**, the folding arm angle being an angle between the push arm **134** and the folding arm **136**. The folding arm angle **146** is determined during operation based on a width of the box blank being expanded, as will be explained later. A drive extension **148** is coupled to the folding arm **136** at or near an end of the folding arm **136** where the folding arm rotation joint **142**, is located. For example, the combination of drive extension **148** and folding arm **136**, as shown in FIG. 3, forms an “L” shape structure. The arm assembly **128** also includes an actuator **150**, for example, a linear pneumatic actuator is shown having a body or cylinder **152** and a piston arm, actuator arm or drive arm **154**. The cylinder **152** is coupled to the base **138** of the arm assembly **128**, using a cylinder tab **156** coupled to the cylinder **152** and a cylinder rotation joint **158** which is secured to the base **138**. In this manner, the cylinder **152** has the freedom to rotate parallel to the X-Y plane as the drive arm **154** extends and retracts. In other embodiments, cylinder **152** can be rotatably secured to arm assembly positioner or **162**. An end of the drive arm **154** is coupled to the drive extension **148** using a pivot point, an rotary joint, or an extension rotation joint **149**, allowing the drive arm **154** and the folding arm **136** to rotate parallel to the X-Y plane. Extension and retraction of the linear actuator **150** is controlled by the machine controller **113**.

(31) Actuator **150** is not limited to a linear actuator. Other examples include a rotation actuator including an electric stepper motor having a rotor coupled to the folding arm and a body or stator coupled to the surface **139** of the base **138**, the rotor acting as the folding arm rotation joint **142**. Another example includes a sheave or pulley coupled to the folding arm centered on the rotation joint **142** and a motor having a stator coupled to the surface **139** of the base **138** and a rotor shaft of the motor coupled to another sheave or pulley, the pulleys coupled together with a belt.

(32) The base **138** of the arm assembly **128** is also coupled to an arm linear track **160**. The arm linear track **160** is parallel to the Y axis. An arm assembly positioner **162** is coupled between the base **138** and the arm linear track **160**.

(33) Examples of an arm assembly positioner **162** may include a linear ball bearing slider mounted to base **138**. A belt running along the arm linear track **160** may be coupled between the linear ball bearing slider and an arm assembly position motor. The arm assembly position motor is electrically coupled to the machine controller **113**. During operation, the machine controller **113** positions the arm assembly **128** along the arm linear track **160**. In another embodiment, the arm assembly positioner **162** may include a motor coupled to the arm linear track to position the arm assembly **128** along the arm linear track **160**.

(34) A side view of the box erecting machine **100** is shown in FIG. 4. The vacuum suction cups **108** are coupled to the gantry **120** above, which is capable of moving parallel to the X axis. The arm assembly **128** is located at a height in the Z direction, at a position, or level, which does not interfere with movement of the arm assembly **128** along the arm linear track **160**. The camera **130** is mounted at a position that provides a camera field of view including a score line of a first box blank **121** in the box blank cartridge **102**. Alternatively, the camera **130** may be adjustably mounted on a camera linear track **170** such that an operator may move the camera **130** horizontally, vertically, diagonally or other direction as indicated by the operator interface **115** to view the first box blank **121**. In addition, the camera **130** is coupled to the camera linear track **170** parallel to the Y axis having a position of the camera **130** determined by a camera servo motor **172** controlled by the machine controller **113**. The camera **130**, in operation, views the score line between a first panel and a second panel of the first box blank **121**. Other distinguishing marks on the box may be used in the camera field of view besides a score line, such as a line or cross hairs printed on the first box blank **121**. The camera **130** is electronically coupled to the machine controller **113** such that an image or series of images may be transferred from the camera **130** to the machine controller **113**.

(35) FIG. 5A is a perspective view of portion of the expanding apparatus **106** of FIG. 1 looking toward the box blank cartridge **102**. The perspective view is from behind and above camera **130**. In the illustrated embodiment, the camera **130** is positioned below the vacuum suction cups **108**

facing toward the first box blank **121** in the box blank cartridge **102**. In the illustrated embodiment, camera **130** has a field of view **174** that includes the score line **176** of the first box blank **121** between the two panels **178** and **180** of the first box blank **121**. The first box blank **121** is being held in position by the gate latch **132** (see FIG. 5A), prior to the vacuum suction cups **108** moving toward the first box blank **121** to pull the first box blank **121** into the expanding apparatus **106**. Examples of camera **130** which are useful in accordance with the present embodiment include video cameras, CCD cameras, CMOS cameras, and machine vision cameras having an on-board processor. The camera is electronically coupled to the machine controller **113** and may communicate an image of the camera's field of view to the machine controller **113** for processing. Alternatively, the camera **130** may process the image and communicate alternate information such as a distance and a direction to the machine controller **113**. Machine controller can control the operation of the camera, including such features as zoom, focus and exposure.

(36) FIG. 5B is an example of the field of view **174** of FIG. 5A. The field of view **174** includes a score line **176** preformed between panels **178** and **180** of the first box blank **121** (see FIG. 5A). A corner of the field of view **174**, such as the lower right corner, may serve as a datum or reference point **182**. The machine controller **113** may be programmed to recognize the score line **176**, determine an actual position of the score line **186** and calculate an actual score line distance **188** from the reference point **182**. Alternatively, a reference line **184** may be used to determine a plurality of distances orthogonally from the reference line **184** to the score line **176**. Because the score line **176** may have a non-zero width, the actual position of the score line **186** may, for example, be an average position of the score line **176**. Other means of determining the actual position of the score line **186** may be used including, for example, a position of one of score line edges **177a** or **177b**, or a weighted average of the positions of score line edges **177a** and **177b**. Because of variations in box blanks caused by, for instance, manufacturers, material or manufacturing location, the determination of the actual score position, may be found empirically.

(37) The actual position of the score line **186** may differ from an expected position of the score line **190**. An expected score line distance **192** may be determined as a difference in position between the reference point **182** and the expected position of the score line **190**. A score line offset **194** may be determined by subtracting the actual score line distance **188** from an expected score line distance **192**. The actual position of the score line **186** may be used, in operation, to position the arm assembly **128** such that the side **140** of the push arm **134** is, for example, aligned with the score line **176** of the first box blank **121**. Alternate positions of the push arm **134** may be used other than alignment with the score line **176**, as is empirically determined to be advantageous in expanding the box blank, for example, a small distance behind the score line toward the second panel.

(38) The actual score line distance **188** may be compared to a set of known score line distances to determine box blank attributes such as a size of the box blank, misalignment of the first box blank **121** or a suitability or fitness of the box blank for expansion. Similarly, the score line offset **194** may be used to determine misalignment of the box blank or the suitability of the box blank for expansion.

(39) To set the field of view **174**, adjustment of a position of camera **130** may be performed by an operator to match a particular box blank type or size. Alternatively, camera **130** position may be determined automatically along the camera linear track **170** using the camera servo motor **172** controlled by the machine controller **113** once a box type has been selected using the operator interface **115**. Other camera **130** positioning schemes may also be used such as a dual axis positioning system controlled by the machine controller **113** that positions the camera **130** in a Y-Z plane to obtain a suitable field of view **174**.

(40) An advantage of using the camera **130** to determine the distance of the score line **176** from the reference point **182** includes an ability to position the arm assembly **128** to facilitate folding of the first box blank **121** into an expanded box **110** (see FIG. 1). As shown in FIG. 5C, the arm assembly **128** may be moved along a Y-axis such that the push arm **134** is positioned to accommodate the

expanded box, avoiding blemishing the box and facilitating proper expansion of the box. For example, the first box blank **121** may be dented during expansion if the push arm **134** is advanced in the direction of the exit conveyor **112** past the score line. Alternatively, the first box blank **121** may not fully expand if the arm assembly **128** is too far away from the score line in a direction opposite the exit conveyor **112** (negative Y direction).

(41) FIGS. **6A-6F** show the operation of the expanding apparatus **106**. FIG. **6A** is an initial position of elements of the expanding apparatus **106**, including the arm assembly **128**, the vacuum suction cups **108**, and the camera **130**. Movement of the elements of the expanding apparatus **106** is regulated, or controlled, by the machine controller **113** according to parameters an operator provides using the machine operator interface **115** (FIG. **1**). The operator may provide parameters, examples of which may include a box type, a box width, and an expected score line distance **192**. Alternatively, these parameters may be stored in the machine controller **113** associated with a box type such that the associated parameters can be automatically accessed once the operator enters the box type. After the operator has loaded the box blanks **104** in the box blank cartridge **102** and moved the cartridge follower **114** into position resting against the last box blank **119**, the gate latch **132** prevents the first box blank **121** from entering the expanding apparatus **106**. Upon the operator starting the box erecting machine **100**, the machine controller **113** moves the gantry **120** and vacuum suction cups **108** toward the box blanks **104**.

(42) The machine controller **113** adjusts a position of the vacuum suction cups **108** in the Y direction such that the vacuum suction cups **108** contact a panel of the first box blank. The arm assembly **128** is positioned in the Y axis such that the folding arm **136** will make contact with a panel of the box blank while the blank is being expanded and the push arm **134** of the arm assembly **128** allows room for the expanded box. The camera **130** is positioned in the Y axis so as to allow a field of view of the folding score line **176** of the box blank which faces toward the camera **130**.

(43) FIG. **6B** shows the expanding apparatus **106** after the machine controller **113** has moved the vacuum suction cups **108** into contact with the box blank. Once the vacuum suction cups **108** are in contact with the box blank, a vacuum is applied to the vacuum suction cups **108** and atmospheric pressure pushes the box blank against the vacuum suction cups **108**. The gate latch **132** may be momentarily released by the machine controller **113**, releasing the box blank from the box blank cartridge **102** and allowing the vacuum suction cups **108** to move the box blank into the expanding apparatus **106** as the gantry **120** moves backward away from the box blank cartridge **102**.

(44) FIG. **6C** shows the expanding apparatus **106** while the machine controller **113** is moving the box blank toward the folding arm **136**. When a panel of the box blank contacts the folding arm **136**, the box blank bends along the vertical score lines of the box blank and the box blank begins to expand.

(45) FIG. **6D** shows the expanding apparatus **106** while the box blank **121** is partially expanded. The vacuum suction cups **108** continue to move in the negative X direction moving the box blank **121** against the folding arm **136**, causing the box blank.

(46) FIG. **6E** shows the vacuum suction cups **108** moving the box blank **121** further in the negative X direction, the cylinder **152** pushing the drive extension **148** which rotates to folding arm **136** against the box blank **121**, aiding in the expansion of box blank **121** into the box **121**. A position of the folding arm **136** may be in parallel with the push arm **134** after rotation of the folding arm **136**. Bottom flaps of the box may have been folded and sealed using known techniques.

(47) FIG. **6F** shows the box **121** in position to be pushed toward the exit conveyor. A side, or panel, of box **121** may be in contact with push arm **134**. Depending on the size of the box, the side of box **121**, in contact with the side of box **121**, may also be in contact with the folding arm **136**. The folding arm **136** which may be parallel with the push arm **134**, thereby able to assist pushing box **121** toward the exit conveyor (see FIG. **6H**).

(48) FIG. **6H** shows the expanding apparatus **106** as the box moves toward the exit conveyor **112**.

Vacuum has been removed from the vacuum suction cups **108** equalizing the atmospheric forces on the box panel and the box is free to move independent of the vacuum suction cups **108**. The push arm **134** of the arm assembly **128** pushes the box toward the exit conveyor **112** as the arm assembly **128** along the +Y direction.

(49) An advantage of using the camera **130** includes the ability to reject a box blank for reasons such as incorrect size, improper position or a box blank defect. FIGS. 7A-7D show an example of rejecting a box blank in accordance with some of the disclosed embodiments. FIG. 7A is a perspective view **702** looking into the expanding apparatus **106**. A box blank **704** is in a position to be pulled into the expanding apparatus **106** from the box blank cartridge **102** (see FIG. 1). A field of view **706** of camera **130** is shown and includes an expected or standard position **708** of a score line **710** and an actual position **712** of the score line **710**. A score line offset **706**, or a distance between the standard position **708** and the actual position **712** of the score line **710** may be determined by the machine controller **113** using the camera **130** as described above or may be determined by the camera **130** itself.

(50) Referring now also to FIG. 7B, in accordance with the present embodiment, machine controller **113** compares a score line offset **714** to a first or high score line offset threshold **716**. The score line offset **714** being greater than the high score line offset threshold **716** may trigger an error condition. In accordance with the present embodiment, machine controller **113** compares the score line offset **714** to a second or low score line offset threshold **718**. The score line offset **714** being less than the score line offset threshold **716** may trigger an error condition. The error condition may indicate box blank **704** does not match a type of the box blank entered by the operator using the operator interface **115**. Other conditions that may indicate an error condition include examples such as an incorrectly loaded box blank, a box blank skewed in the carriage, the last box blank has been expanded and the carriage is empty or a damaged box blank. The high and low score line offset thresholds **716** and **718** may be centered around zero or have other values as needed.

(51) Actions may be taken when the score line offset **714** exceeds a threshold. In an alternative embodiment, FIG. 7C shows a back view **720**, a front view **722** and a side view **724** of a set of two vacuum suction cups **108** coupled to a bracket **726** forming components of a box blank rejection mechanism. In side view **728**, the bracket **726** is coupled to a gantry **120** using a rotor **730** controlled by the machine controller **113**, that allows the bracket and vacuum suction cups **108** to rotate in a degree of freedom such as Φ .sub.Y. When the machine controller **113** determines the score line offset **714** exceeds an offset threshold, a box blank rejection action such as that shown in FIG. 7D may occur. In operation, the vacuum suction cups **108** advance toward the box blank **704**, the box blank **704** is pushed against the vacuum suction cups **108** by atmospheric pressure when vacuum is applied to the vacuum suction cups, the gate latch **132** momentarily releases and allows the box blank **704** into the expanding apparatus **106**, the rotor **730** rotates the vacuum suction cups **108** and the box blank **704** over a top of the box blank cartridge **102** of the box expanding machine **100** and releases the box blank **704** over the box blank cartridge **102**. The box erecting machine **100** then proceeds to a next box blank.

(52) FIG. 8 is a flow chart showing a setup operation of the box erecting machine **100** according to an embodiment of the present disclosure. The setup operation begins at a step **802**. In a step **804** an operator selects a box blank type from a menu on an operator interface **115** of a machine controller **113**. The machine controller **113** associates stored parameters specific to the box blank type such as a box blank width, a score line position relative to an edge of the box blank, and an offset threshold. For instance, based on the blank width, the machine controller **113**, or operator, may adjust a right box blank guide **116** of the box blank cartridge **102** to accommodate the box blank width. In step **804**, the operator pulls a cartridge follower **114** back and loads the box blanks **104** into the box blank cartridge **102** such that a score line between box panels is vertical and an edge of the box blank is against the left box blank guide **118**. The cartridge follower **114** is then moved to come into contact with a last box blank **119** in the box blank cartridge **102**. A first box blank **121** is

up against a gate latch **132**, which prevents the first box blank **121** from entering an expanding apparatus **106**. In step **806**, the operator indicates to the machine controller **113** that the box blanks **104** are loaded, and the controller moves an arm assembly **128** to a position along an arm linear track **160** parallel to a Y axis. The position of the arm assembly **128** may be a parameter associated with the box blank type as accessed by the machine controller **113**, referred to as an arm assembly **128** initial position. The arm assembly **128** initial position may be such that a front face of a push arm **134** is aligned with an average, or typical, position of a score line between two panels of the box blanks **104** in the box blank cartridge **102**. Other arm assembly **128** initial positions may be used as determined to be needed for box expansion. In step **808**, the operator aligns a camera **130** to a position in which the camera field of view includes the score line between two panels of the first box blank **121**. The machine controller **113** may examine an image from the camera to aid the operator in the alignment of the camera **130** using, for example, an audible sound from the operator interface **115**. Alternatively, the camera **130** may ride on an arm linear track **160** parallel to the Y axis having a position encoder, and the machine controller **113** may automatically position the camera **130** along the linear track using a camera **130** position actuator based on a position parameter associated with the box blank type. The camera **130** position may be such that a distance between a reference point **182** in the camera field of view **174** and the score line **176** of the box blank is an average offset **186** for the box blank type. In step **810**, the machine controller **113** uses the pneumatic actuator of the arm assembly **128** to adjust an angle between a front face of a folding arm **136** and a front face of the push arm **134** to a value of a stored folding arm angle parameter specific to the box type. The setup operation of the box erector machine ends with step **812**.

(53) FIG. **9** is a flow chart **900** including steps for erecting a box from a box blank after the setup operation **800** is performed. The box erecting process starts at step **902**. In step **904**, the camera images a field of view that is intended to include a box blank feature, such as a score line between a first and second panel of a next box blank in the box blank cartridge **102**. Other box blank features may include a mark, a line or a pattern printed on the box. The image is electronically transmitted from the camera **130** to the machine controller **113**. The machine controller **113** first examines the image to determine if the next box blank is present in decision **906**. If the next box blank is present (yes) then the box erecting procedure **900** goes to step **908**, in which the machine controller **113** determines an actual distance between the score line and a reference point and a score line for the next box blank. The distance between the reference point and a score line may be an actual position of the score line referenced to the reference point in the image. Proceeding to step **909**, an offset may be determined by subtracting the distance between the score line and the reference point from an expected distance between the scoreline and the reference point. Alternatively, the offset may be determined by using the reference point and the expected distance between the score line and the reference point to determine an expected position of the score line, from which the actual position of the score is be subtracted.

(54) The box erecting process proceeds to a decision step **910**, to determine whether the score line offset **194** exceeds an offset threshold. Examples of the offset threshold may include a simple plus and minus offset threshold or a more complex offset threshold such as a plus offset threshold and a minus offset threshold which have non-equal magnitudes. Examples of situations in which the score line offset **194** may exceed the offset threshold include the next box blank being the wrong size, or the wrong box blank type. Alternatively, the machine controller **113** may use a reference line **184**, such as an edge of the image and determine if any one or more points along the score line exceed an offset threshold. Examples of a situation in which the score line offset **194** may exceed an offset threshold along the score line may include a box blank that is at an angle, or improperly loaded, into the box blank cartridge **102**. If an answer to the decision “Does the score line offset **194** exceed the offset threshold” is “Yes” then the process proceeds to step **1000**, which is detailed in FIG. **10**, and the box blank is ejected. Alternatively, operation of the box erecting machine **100** may stop and alert the operator to the error. After the box blank is ejected, then the box erecting

process proceeds back to step **904**. If the answer to decision step **910** “Does the score line offset **194** exceed the offset threshold” is “No” then the process **900** proceeds to step **912**. In step **912**, the arm assembly **128** moves to align a front face of the push arm **134** of the arm assembly **128** with the score line of the next box blank using the score line offset **194** determined in step **908**. For example, if the score line offset **194** of a box blank is 1 inch to the right of the average score line offset **194**, then the arm assembly **128** may move 1 inch to the right from the arm assembly **128** initial position.

(55) After step **912**, the process **900** moves to step **913** in which the controller determines a folding arm angle associated with the box blank type being expanded. In step **914** the controller positions, or rotates, the folding arm to the determined folding arm angle. The process **900** may move to step **915**, in which a gantry **120** moves toward the box blank cartridge **102**, moving vacuum suction cups **108** toward the next box blank until the vacuum suction cups **108** contact the first panel of the next box blank. Next, in step **916**, vacuum is applied to the vacuum suction cups **108**, and atmospheric pressure presses the next box blank against the vacuum suction cups **108** allowing the vacuum suction cups **108** to exert force onto the first panel of the next box blank. In step **918**, the gate latch **132** momentarily retracts allowing the next box blank to be pulled into the expanding apparatus **106** by the vacuum suction cups **108** and the gantry **120**. In step **920**, as the vacuum suction cups **108** are pulling the next box blank into the expanding apparatus **106**, the second panel of the box blank contacts the folding arm **136** and the box blank bends at score lines and the box begins to expand or open. In step **922**, the box blank continues to expand as the second panel of the box blank is pulled against the folding arm **136**. In an embodiment, the folding arm angle, or the angle of the folding arm **136** to the push arm **134**, may stay constant during the expanding process. Alternatively, the linear actuator **150** of the arm assembly **128** may extend the drive extension **148** to reduce the folding arm **136** angle, aiding in the folding at the score line between the first and second panel of the box blank.

(56) In step **924**, the box blank stop moving the box blank at some time after the box has expanded and in step **926** the vacuum is released from the vacuum suction cups **108**, releasing the box from the vacuum suction cups **108**. In step **928** the arm assembly **128** moves the box toward the exit conveyor **112** by pushing the box against a surface of the push bar. After the box is pushed out of the expanding apparatus **106**, in step **930** the arm assembly **128** may return to its initial position along the arm linear track **160** and the box expanding process returns to step **904**, in which the camera **130** images the field of view.

(57) Returning to decision step **906**, if the answer to the question “Next box blank is present?” is “No” then the box expanding process **900** ends at step **932**.

(58) FIG. **10** shows the box blank ejection process **1000** being initiated at step **1002**. In step **1004**, the gantry **120** moves the vacuum suction cups **108** toward the next box blank and contacts the next box blank. In step **1006**, vacuum is applied to the vacuum suction cups **108**, and atmospheric pressure pushes the next box blank **121** against the vacuum suction cups **108**. Next, in step **1008**, the gate latch **132** momentarily retracts allowing the vacuum suction cups **108** to pull the next box blank **121** into the expanding apparatus **106**. In step **1010**, the arm assembly **128** moves on the arm linear track **160** away from the box blank **121** to ensure the folding arm **136** does not contact the box blank **121**. Process **900** moves to step **1012** in which the vacuum suction cups **108** are rotated by rotor **730** (see FIG. **7C**) by an angle $\Phi_{sub.Y}$, lifting the box blank and clearing the box blanks **104** in the box blank cartridge **102**. In step **1014**, the controller **113** releases the vacuum from the vacuum suction cups **108**. In step **1016** gravity causes the box blank to fall over the box blank cartridge. Alternatively, the gantry may move the box blank toward the blank cartridge while the vacuum is released, giving the box blank some momentum to clear the box blank cartridge. In step **1118**, the vacuum suction cups are rotated by the rotor **730** by an angle which may be the negative of $\Phi_{sub.Y}$, the gantry **120** and vacuum suction cups **108** returning to a position, preferably out of the camera field of view **174**, and the box blank ejection process **1000** ends at step **1020**.

(59) Alternate methods of ejecting a box blank may be used, including notifying an operator that a box blank needs to be removed from the expanding apparatus **106** or rotating the vacuum suction cups by an angle such that the box blank is ejected out an alternate side of the box erecting machine **100**.

(60) Embodiments in accordance with the present disclosure include an apparatus for expanding a box blank. Some described embodiments include an arm assembly, a controller and a box blank conveyor. The arm assembly includes a base having a surface, a first arm coupled to the surface of the base through a rotation joint, the first arm having a surface perpendicular to the surface of the base, and an actuator. The actuator includes a body, coupled to the surface of the base, and an actuator arm coupled to the first arm, a position of the actuator arm determining a rotation angle of the first arm around the rotation joint. The controller is coupled to the actuator of the arm assembly. The apparatus includes a box blank conveyor, which, in operation, moves the box blank against the surface of the first arm of the arm assembly, expanding the box blank. A controller is present which, in operation, sets the rotation angle of the first arm around the rotation joint based on a first dimension of the box blank.

(61) In other embodiments, the present disclosure includes an apparatus, including a controller, a push extension a folding extension, a box blank conveyor and a camera. The push extension, in operation, is positioned by the controller in a first direction. The folding extension is rotationally coupled to the push extension. The folding extension, in operation, forms an angle with the push extension, the angle determined by the controller. The apparatus also includes a box blank conveyor, coupled to the controller, which, in operation, moves a box blank in a second direction different from the first direction to contact the folding extension. The apparatus also includes a camera electronically coupled to the controller, the camera having a field of view during operation which includes a feature of the box blank, a position of the feature in the field of view being determined by the camera or the controller. The controller, in operation, uses the position of the feature in the field of view to position the push extension in the first direction, such that, in operation, the box blank conveyor moves the box blank in the second direction to contact the folding extension and cause the box blank to expand.

(62) Other embodiments of the present disclosure, relate to methods for expanding a box blank, including imaging a box blank using a camera having a field of view including a feature of the box blank to generate an image of the field of view. The methods include determining a position of the feature of the box blank using the image of the field of view. The method includes moving a folding arm in a first direction to align the folding arm with a panel of the box blank using the position of the feature of the box blank. The method determines a folding arm angle using the position of the feature of the box blank and rotates the folding arm to the folding arm angle by a folding arm actuator coupled to the controller. The method also includes moving the box blank in a second direction, different from the first direction, by a box blank conveyor to contact the folding arm.

(63) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An apparatus for expanding a box blank, comprising: an arm assembly, including: a base having a surface; a first arm coupled to the surface of the base through a rotation joint, the first arm having a box blank contact surface perpendicular to the surface of the base; and an actuator, including: a body, coupled to the surface of the base; and an actuator arm coupled to the first arm; a box blank conveyor, which, in operation, moves the box blank against the box blank contact surface of the first arm of the arm assembly, expanding the box blank; a camera having a field of view including a feature of the box blank; a controller electrically coupled to the camera, the controller in operation, receives an image from the camera, determines a position of the feature of the box blank in the field of view and moves the arm assembly in a first direction using the actuator based on the position of the feature in the field of view; and a box blank rejection mechanism, the box blank rejection mechanism including at least one suction cup coupled to a bracket, the bracket coupled to a gantry via a rotor.
2. The apparatus of claim 1, the box blank conveyor including: a vacuum cup; a valve to selectively couple the vacuum cup to a vacuum; a positioner coupled to the vacuum cup; and the controller coupled to the positioner and the valve, the controller, in operation, causes the positioner to move the vacuum cup against the box blank, opens the valve to couple the vacuum to the vacuum cup, and causes the box blank to move in a second direction against the box blank contact surface of the first arm.
3. The apparatus of claim 1, further comprising a first linear actuator coupled to the arm assembly, the controller, in operation, causes the linear actuator to move the arm assembly in a second direction.
4. The apparatus of claim 3, wherein the first direction is transverse to the second direction.
5. The apparatus of claim 4, wherein the first linear actuator coupled to the arm assembly includes: a linear track; and a positioner coupled between the linear track and base of the arm assembly.
6. The apparatus of claim 1, wherein the rotor is controlled by a feature of the box blank includes a score line of the box blank and the position of the feature is relative to a reference point.
7. The apparatus of claim 1, wherein the controller electrically coupled to the camera, in operation, upon receiving the image from the camera and determining the position of the feature of the box blank in the field of view, rotates the first arm of the arm assembly to a rotation angle based on the position of the feature of the box blank in the field of view.
8. An apparatus, comprising: a controller; a push extension positionable, by the controller during operation, in a first direction; a folding extension positionable, by the controller, during operation, the folding extension in operation forming an angle with the push extension the angle determinable by the controller; a box blank conveyor, which in operation moves a box blank including a feature in a second direction different from the first direction to contact the folding extension; a first linear actuator coupled to the push extension; and a box blank rejection mechanism, the box blank rejection mechanism including at least one suction cup coupled to a bracket, the bracket coupled to a gantry via a rotor, the controller, in operation, sets a rotation angle of the folding extension around a rotation joint based on the position of the feature.
9. The apparatus of claim 8, further comprising: a base coupled between the push extension and the folding extension and the rotation joint coupled between the base and the folding extension, which allows the folding extension to rotate; and an actuator coupled to the folding extension, the actuator having a position determined by the controller, which, in operation, uses the position of the feature to determine the position of the actuator.
10. The apparatus of claim 9, the actuator being a linear actuator coupled between the base and the folding extension.
11. The apparatus of claim 10, the folding extension including a drive extension, the drive extension coupled between the linear actuator and the folding extension.
12. The apparatus of claim 8, the box blank conveyor including: a vacuum suction cup, selectively

coupled to a vacuum; an actuator, which, in operation, positions the vacuum suction cup in the second direction; and a bracket coupled between the actuator and the vacuum suction cup.

13. The apparatus of claim 8, further comprising a camera electronically coupled to the controller, the camera having a field of view, which, during operation, includes the feature of the box blank, the position of the feature in the field of view being determined by the camera or the controller.

14. The apparatus of claim 13, wherein the feature of the box blank in the field of view is a score line.

15. A method for expanding a box blank, including: determining a position of a feature of the box blank relative to a reference point; moving, based on the position of the feature of the box blank relative to a reference point, a folding arm in a first direction to align the folding arm with a panel of the box blank; determining, using the position of the feature of the box blank relative to reference point, a folding arm angle; rotating the folding arm to the folding arm angle; and moving the box blank in a second direction, different from the first direction causing the box blank to contact the folding arm.

16. The method of claim 15, further including: rotating the folding arm toward the box blank by the folding arm actuator while the folding arm is in contact with the box blank.

17. The method of claim 15, further including: rejecting the box blank by a box blank rejection mechanism if an offset of the position of the feature of the box blank exceeds a threshold.

18. The method of claim 17, wherein rejecting the box blank by a box blank rejection mechanism further includes contacting the box blank with a suction cup of the box blank rejection mechanism.

19. The method of claim 17, wherein the rejecting the box blank by a box blank rejection mechanism is based on the position of the feature of the box blank relative to the reference point.

20. The method of claim 15, further including rejecting the box blank by lifting the box blank over a box blank cassette.
